Item 13: negation-map shortfall and ECC2K-130 plan

Run report and result. Protocol `PROTOCOL.md` (frozen 2026-10-18 before any measured run); cards 01 to 15; instrument `examples/blindspots_13_rho_negation.rs`; read-out `raw/summary.md`. Host: shared 4-vCPU container, isolation L0, one thread per job. All numbers below are measured or derived from the counted operations; derivations are in the README and the protocol.

Verdict

TRADES (level: constant factor) for the negation walk against the un-quotiented Teske walk; **DELIVERED** for the part B plan. Frozen first prediction: INCONCLUSIVE (11, 14, 12 collision-preserving cells reach 1.35 at 2^40, 2^44, 2^48 under free native work, none robustly). Second prediction: a sign change exists in the tested range (12 settings are clear net losses at all three sizes). Per-setting: 0 ADVANCES, 30 TRADES, 2 MATCHES, 12 REGRESSES. Every gate passed: baseline against the ideal-map control (0.948 [0.882, 1.019], 1.023 [0.943, 1.104], 1.008 [0.932, 1.087]), ideal-map control sees sqrt 2 (1.490, 1.412, 1.409), no-escape absorbed at r = 16, A/A covers 1 in 12 of 12 cells.

Gain over the plain walk

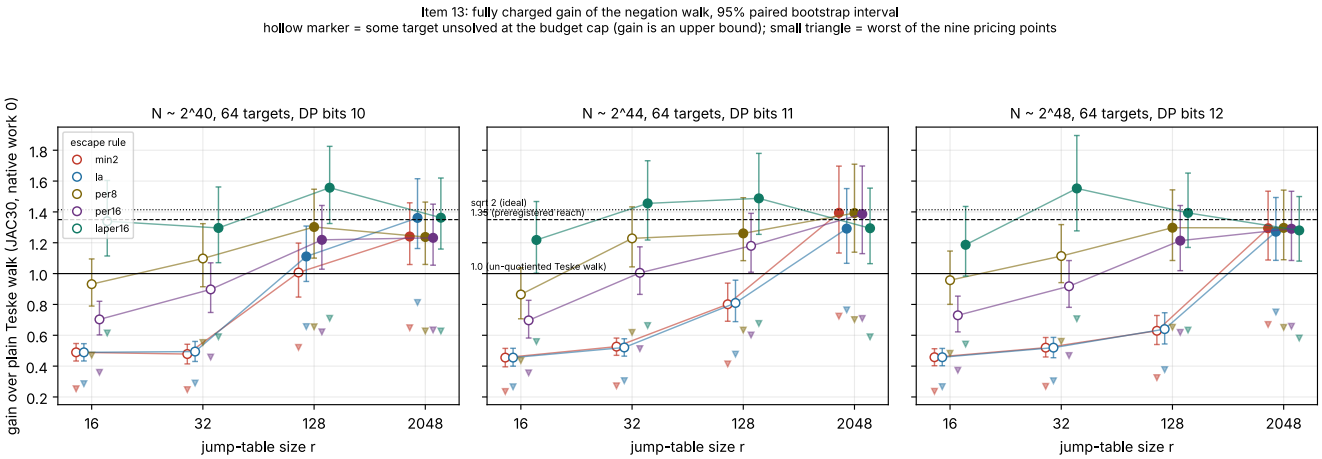


Figure 1: Fully charged gain (Jacobian model, native work 0), 64 paired targets per cell, 95% paired percentile bootstrap (4,000 resamples). Hollow markers: at least one target unsolved at the cap, gain is an upper bound. Small triangles: worst of the nine pricing points. Source: `raw/summary.json`, drawn by `plot_graph.py`. Item 13 sweeps, seed offsets 7000000 and 9000011.

Distance remaining. Gain 1.0 is rho (the plain walk); the target is sqrt 2 = 1.414. Best collision-preserving cell per size, native work free: 1.557 [1.325, 1.825] (2^40, `laper16`, r = 128), 1.488 [1.254, 1.779] (2^44, same), 1.552 [1.277, 1.895] (2^48, `laper16`, r = 32), i.e. 1.10, 1.05, 1.10 of sqrt 2 with intervals that include 1.35 and sqrt 2. At the harshest pricing point (native operations at one group addition each, Jacobian) every collision-preserving cell is below 1.0 (nine-point minimum 0.21 to 0.81). The order is unresolved across the section 5a band, which is what TRADES records.

Every collision-preserving setting

Point gain and 95% interval, fully charged JAC30, native work 0; + marks a censored cell (upper bound). The nine-point range is the minimum and maximum point gain over JAC30, AFF3, AFF30 times native work 0, 0.1, 1 over the three sizes. `best/sqrt2` is the largest of the three point gains divided by 1.414.

rule	r	2^40	2^44	2^48	nine-point range	best/sqrt2	verdict
min2	16	0.49 [0.43, 0.55]+	0.46 [0.40, 0.51]+	0.46 [0.40, 0.51]+	0.24..0.51	0.35	REGRESSES
la	16	0.49 [0.43, 0.55]+	0.46 [0.40, 0.52]+	0.46 [0.40, 0.51]+	0.27..0.51	0.35	REGRESSES
per2	16	0.49 [0.43, 0.55]+	0.46 [0.39, 0.52]+	0.46 [0.40, 0.51]+	0.21..0.51	0.35	REGRESSES
per4	16	0.76 [0.65, 0.88]+	0.63 [0.52, 0.74]+	0.54 [0.47, 0.62]+	0.26..0.80	0.54	REGRESSES
per8	16	0.93 [0.79, 1.09]+	0.86 [0.71, 1.04]+	0.96 [0.80, 1.15]+	0.44..1.00	0.68	MATCHES
per16	16	0.70 [0.60, 0.82]+	0.70 [0.58, 0.83]+	0.73 [0.62, 0.85]+	0.36..0.76	0.52	REGRESSES
per32	16	0.55 [0.48, 0.63]+	0.57 [0.48, 0.66]+	0.56 [0.48, 0.65]+	0.28..0.59	0.40	REGRESSES
per64	16	0.51 [0.45, 0.57]+	0.50 [0.44, 0.57]+	0.49 [0.42, 0.56]+	0.25..0.53	0.36	REGRESSES
laper16	16	1.34 [1.11, 1.60]+	1.22 [1.00, 1.47]	1.19 [0.98, 1.44]	0.54..1.40	0.95	TRADES
laper32	16	1.27 [1.06, 1.51]+	1.15 [0.95, 1.38]	1.12 [0.94, 1.36]	0.52..1.33	0.90	TRADES
laper64	16	1.14 [0.96, 1.37]+	1.04 [0.85, 1.23]	1.01 [0.85, 1.22]+	0.47..1.20	0.81	TRADES
min2	32	0.48 [0.41, 0.54]+	0.53 [0.47, 0.58]+	0.52 [0.46, 0.59]+	0.25..0.55	0.37	REGRESSES
la	32	0.49 [0.43, 0.56]+	0.52 [0.46, 0.58]+	0.52 [0.45, 0.59]+	0.29..0.55	0.37	REGRESSES
per2	32	0.48 [0.41, 0.54]+	0.53 [0.47, 0.58]+	0.52 [0.46, 0.58]+	0.22..0.55	0.37	REGRESSES
per4	32	1.19 [0.98, 1.44]+	1.28 [1.09, 1.49]+	1.10 [0.93, 1.30]+	0.54..1.34	0.90	TRADES
per8	32	1.10 [0.91, 1.32]+	1.23 [1.04, 1.43]+	1.11 [0.94, 1.32]+	0.55..1.28	0.87	TRADES
per16	32	0.90 [0.75, 1.07]+	1.00 [0.87, 1.17]+	0.92 [0.78, 1.08]+	0.46..1.05	0.71	TRADES
per32	32	0.68 [0.58, 0.81]+	0.76 [0.66, 0.88]+	0.72 [0.61, 0.83]+	0.35..0.80	0.54	REGRESSES
per64	32	0.55 [0.47, 0.63]+	0.61 [0.54, 0.69]+	0.59 [0.51, 0.68]+	0.28..0.64	0.43	REGRESSES
laper16	32	1.30 [1.07, 1.56]	1.46 [1.22, 1.73]	1.55 [1.28, 1.90]	0.59..1.62	1.10	TRADES
laper32	32	1.28 [1.05, 1.55]+	1.44 [1.20, 1.71]	1.53 [1.26, 1.88]	0.59..1.60	1.08	TRADES
laper64	32	1.25 [1.03, 1.51]+	1.40 [1.17, 1.67]	1.49 [1.22, 1.83]	0.57..1.56	1.05	TRADES

min2	128	1.01 [0.85, 1.20]+	0.80 [0.69, 0.94]+	0.63 [0.54, 0.73]+	0.33..1.05	0.71	TRADES
la	128	1.11 [0.95, 1.31]	0.81 [0.69, 0.96]+	0.64 [0.54, 0.75]+	0.38..1.16	0.79	TRADES
per2	128	1.00 [0.85, 1.18]+	0.80 [0.69, 0.94]+	0.63 [0.54, 0.73]+	0.29..1.05	0.71	TRADES
per4	128	1.35 [1.13, 1.60]	1.30 [1.12, 1.54]	1.34 [1.12, 1.59]	0.64..1.40	0.95	TRADES
per8	128	1.30 [1.10, 1.55]	1.26 [1.08, 1.49]	1.30 [1.08, 1.54]	0.63..1.36	0.92	TRADES
per16	128	1.22 [1.03, 1.44]	1.18 [1.01, 1.39]+	1.21 [1.02, 1.44]	0.60..1.27	0.86	TRADES
per32	128	1.08 [0.91, 1.27]+	1.05 [0.90, 1.23]+	1.07 [0.89, 1.27]+	0.54..1.13	0.76	TRADES
per64	128	0.88 [0.75, 1.04]+	0.87 [0.75, 1.01]+	0.88 [0.75, 1.05]+	0.45..0.93	0.63	MATCHES
laper16	128	1.56 [1.32, 1.83]	1.49 [1.25, 1.78]	1.39 [1.17, 1.65]	0.63..1.62	1.10	TRADES
laper32	128	1.56 [1.32, 1.83]	1.49 [1.25, 1.77]	1.39 [1.17, 1.66]	0.64..1.62	1.10	TRADES
laper64	128	1.55 [1.33, 1.83]	1.48 [1.25, 1.77]	1.39 [1.16, 1.66]	0.64..1.62	1.10	TRADES
min2	2048	1.24 [1.06, 1.46]	1.39 [1.13, 1.70]	1.29 [1.09, 1.53]	0.65..1.44	0.99	TRADES
la	2048	1.36 [1.16, 1.62]	1.29 [1.07, 1.55]	1.27 [1.09, 1.49]	0.75..1.38	0.96	TRADES
per2	2048	1.24 [1.05, 1.45]	1.39 [1.13, 1.72]	1.29 [1.09, 1.55]	0.58..1.44	0.99	TRADES
per4	2048	1.24 [1.06, 1.47]	1.40 [1.12, 1.72]	1.30 [1.10, 1.54]	0.61..1.44	0.99	TRADES
per8	2048	1.24 [1.06, 1.46]	1.39 [1.14, 1.71]	1.30 [1.09, 1.54]	0.63..1.44	0.98	TRADES
per16	2048	1.23 [1.05, 1.45]	1.39 [1.13, 1.70]	1.29 [1.09, 1.53]	0.63..1.43	0.98	TRADES
per32	2048	1.22 [1.05, 1.44]	1.37 [1.11, 1.69]	1.28 [1.07, 1.52]	0.63..1.42	0.97	TRADES
per64	2048	1.20 [1.03, 1.41]	1.35 [1.09, 1.66]	1.26 [1.06, 1.49]	0.62..1.40	0.95	TRADES
laper16	2048	1.36 [1.16, 1.62]	1.29 [1.06, 1.55]	1.28 [1.08, 1.50]	0.58..1.38	0.96	TRADES
laper32	2048	1.36 [1.17, 1.62]	1.29 [1.07, 1.55]	1.28 [1.09, 1.50]	0.59..1.38	0.96	TRADES
laper64	2048	1.36 [1.16, 1.61]	1.29 [1.07, 1.55]	1.28 [1.09, 1.50]	0.59..1.38	0.96	TRADES

Controls, extension, caveats

Selftest (card 01): two-implementation agreement at 2^20, positive controls, no wrong logarithm. Fruitless 2-cycle entry rate per draw against 1/(2r): 48 settings, ratio 0.948 to 1.039 (HEUR-RHO-1 not refuted). Size dependence: 4 of 44 collision-preserving pairs have disjoint intervals at 2^40 and 2^48 (`la`, `min2`, `per2` at r = 128; `per4` at r = 16). Theta extension at 2^44, r = 128 (outside the frozen rule): `la` 1.344 / 0.810 / 0.443 at DP bits 8 / 11 / 14, `laper16` 1.474 / 1.488 / 1.179, so the size dependence is a DP-length dependence. Caveats: toy scale 2^40 to 2^48 in a counted `Z/N` model (curve agreement only at 2^20); not an ECC2K-130 or NIST measurement; the 1 GAE point is a harsh bound; no exponent is touched. Relay check: 9 of 15 executor reports verbatim, 6 deviated in format (card 12 alters one hex digit of the card sha256); receipts all completed and 33 output digests match disk (`raw/relay-check.txt`).

Part B: costed ECC2K-130 plan

Measured: current `ecc2k130/` client, `--curve 131 --bench --threads 1`, five runs 14.702, 15.189, 14.532, 15.206, 15.605 M iterations/s (median 15.189, min 14.532, max 15.605; load average 0.83 before, 1.45 after; L0). Cited, not re-measured: GPU rates 15.884369 B/s (RTX PRO 6000 sigma-table) and 12.499203 B/s (RTX PRO 4500 x-only bridge, work factor 1.006696), 2^60.9 iterations, DP intervals 2^28.41 and 2^25.27, 2026-09 prices.

quantity	**value**
expected work	2.1514e18 iterations
CPU alone	3.935e7 core-hours = 4,488 core-years
DPs and storage at 32 B (w32 / w34)	6.03e9, 193 GB / 5.32e10, 1.70 TB
RTX PRO 6000 sigma-table	37,623 GPU-h; $45,629 at $1.2128/h; $155,760 at $4.14/h on-demand
RTX PRO 4500 x-only (other iteration function)	48,133 GPU-h; $27,460 at $0.5705/h
50% / 90% / 99% (Rayleigh model)	35,345 / 64,419 / 91,103 GPU-h
wall clock for 100 / 1,000 / 10,000 GPUs	15.7 / 1.6 / 0.2 days
break-even CPU core-hour price	$0.00116

No live price, no GPU rate measured here, no campaign progress credit, no network or orchestration cost. Sources: `raw/plan.md`.